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WEEK-END PRE-PROJECT PRESENTATION DATA SHEET

Domain-wise project calibration / evaluation framework. This helps evaluate whether a student **actually understands the project they built.**

For **Data Analytics, MERN Full Stack, and AI-ML projects.**

STUDENT NAME: Aditya Patil

DOMAIN NAME: Data Analytics

PROJECT NAME: Student Performance Analysis in STEM Subjects

1. Data Analytics Project Calibration Points

S. No.	Calibration Point	What Evaluator Checks	Example	Student Remark
1	Problem Statement	What business or data problem is being solved?	Sales performance analysis	The project analyzes student performance in STEM subjects to identify factors affecting academic scores and understand patterns in student learning outcomes.
2	Data Source	Where the dataset came from	Kaggle, company database, API	Dataset was collected from Kaggle containing student academic records, demographic details, study habits, and subject-wise performance data.
3	Data Size & Structure	Rows, columns, type of data	50k rows, 12 columns	1k rows, 7 columns
4	Data Cleaning	Handling null values, duplicates, outliers	Missing values replaced	Removed duplicate records, handled missing values, corrected inconsistent entries, and standardized column formats for analysis.
5	Data Transformation	Feature engineering, normalization	Created revenue column	Created performance categories such as high, medium, and low performers, and transformed categorical variables for better analysis.

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S. No.	Calibration Point	What Evaluator Checks	Example	Student Remark
6	Exploratory Data Analysis (EDA)	Patterns, correlations identified	Region vs Sales trend	Analyzed relationships between study hours, attendance, parental education, and student scores in mathematics and science subjects.
7	Tools Used	Why those tools were selected	Python, SQL, Power BI	Python was used for data preprocessing and analysis, Pandas and NumPy for handling datasets, and Matplotlib/Seaborn for visualizations.
8	Visualization	Quality of dashboards/charts	Interactive Power BI dashboard	Created bar charts, histograms, correlation heatmaps, and scatter plots to visualize academic performance trends and influencing factors.
9	Insights Generated	Key insights from data	Q4 sales drop in north region	Students with higher study hours generally scored better in STEM subjects. Attendance and parental education also showed a positive impact on academic performance.
10	Business Impact	How insights help business	Inventory optimization	The insights can help schools and educational institutions improve student performance strategies and identify students needing additional academic support.
11	Automation	Any automation done	Scheduled data refresh	Automated preprocessing and analysis workflow using Python scripts in Google Colab for faster data analysis.
12	Limitations	Dataset limitations	Incomplete historical data	Dataset size was limited and did not include real-time student monitoring or psychological factors affecting performance.
13	Future Scope	Improvements possible	Real-time analytics	Future improvements include machine learning models for predicting student performance and interactive dashboards for educational analytics.

Universal Questions to Check if the Candidate Really Built the Project

Question	Purpose	Student Remark
Why did you choose this problem?	Checks understanding of problem	I chose this project because education analytics is an important field, and I wanted to

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Question	Purpose	Student Remark
		understand how different factors influence student performance in STEM subjects.
What challenges did you face?	Checks real development experience	The main challenges were handling categorical data, cleaning inconsistent records, and identifying meaningful correlations between academic factors and student scores.
What would you improve if given more time?	Checks depth of thinking	I would build a predictive machine learning model to forecast student performance and create a more interactive dashboard for teachers and institutions.
What alternative technologies could be used?	Checks technical maturity	Instead of Python and Colab, technologies like Tableau, Power BI, Apache Spark, or SQL-based analytics systems could also be used.
How would you scale this to 1 million users?	Checks system design thinking	I would use cloud-based databases, distributed data processing systems, and scalable ETL pipelines to efficiently process and analyze large educational datasets.

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